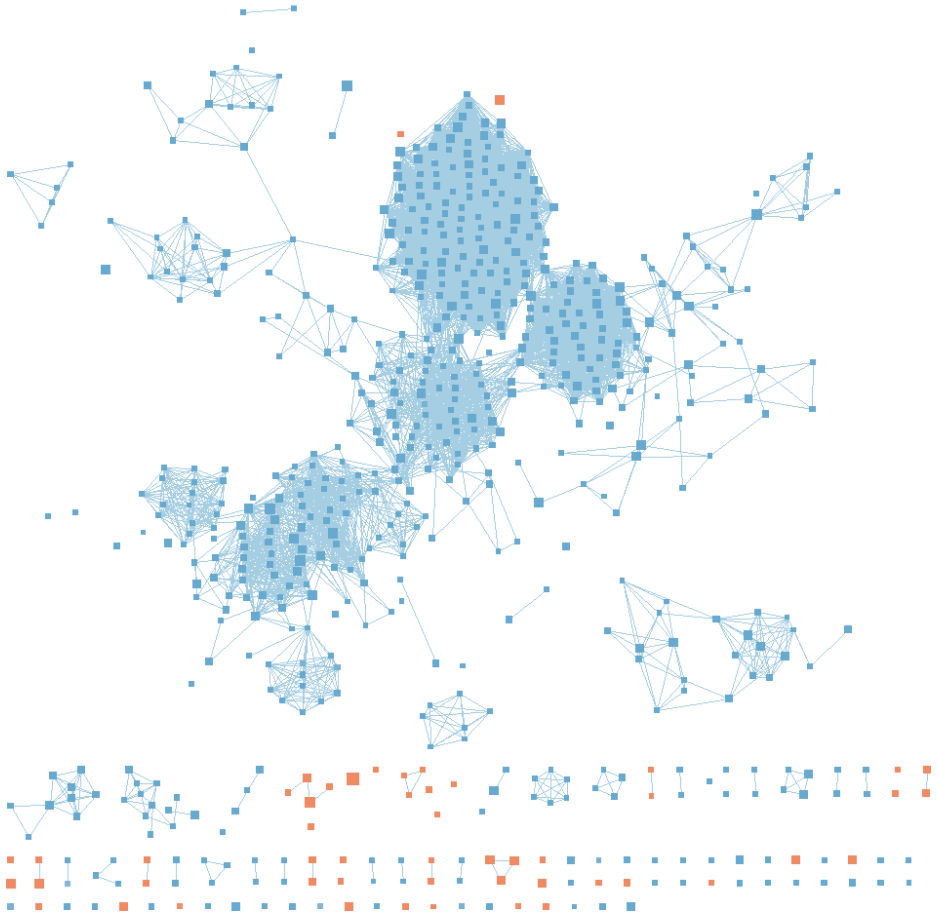

Assignment 3: Presentation of Pathway and Network Analysis Results

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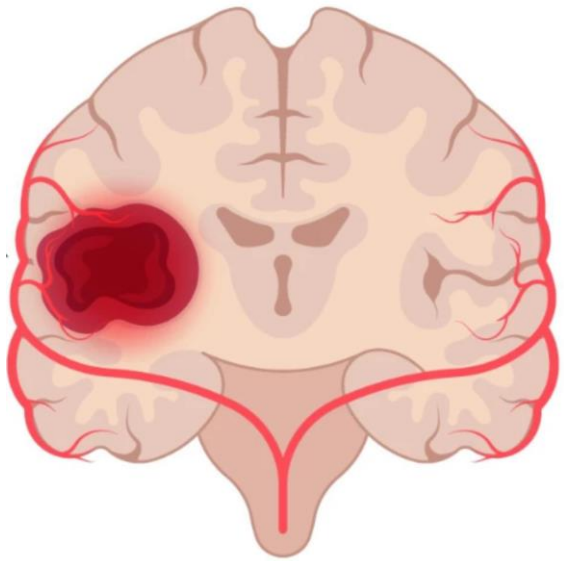
March 24, 2026

BCB420H1: Computational Systems Biology

Intro to dataset and motivation



Glioblastoma (GBM)

- Aggressive form of brain cancer
  - Characterised by its stemness
  - ↑ tumour growth & recurrence
- Treatment option: TMZ-RT (temozolomide + radiation)
- Poor survival prognosis, even after treatment

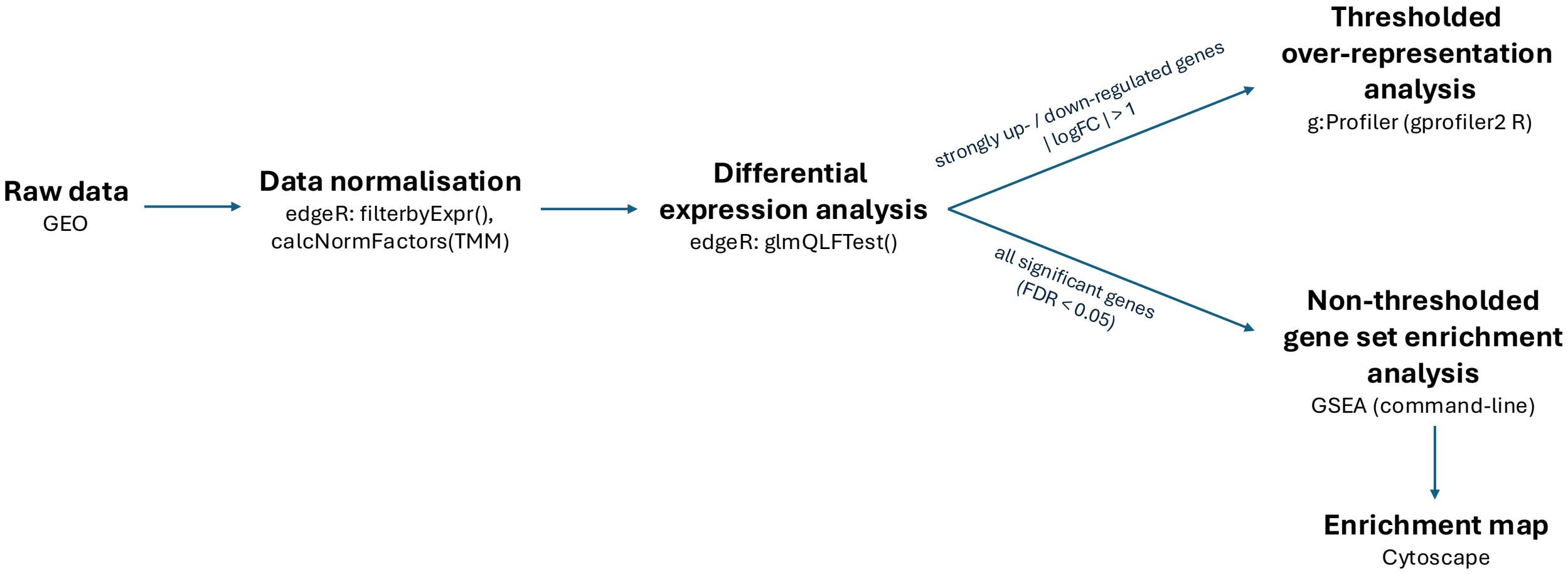
GSE253464 (Bulk RNA-seq data)

- Disease stage-specific role of the mitochondrial pyruvate carrier suppresses differentiation in temozolomide and radiation-treated glioblastoma (Martell et al., 2025)
- Investigate role of the MPC (mitochondrial pyruvate carrier) in GBM
- Added a metabolic inhibitor of the MPC (UK-5099)
  - In TMZ-RT treated GBM-cells
- 3 naïve samples, 3 treated samples

Personal motivation

- BCB430 project: Reconstructing glioblastoma tumour lineages with single-cell DNA methylation data

Method pipeline



Normalisation of dataset

Filtering lowly expressed genes

```
# Create count-only matrix
cleaned_data_matrix <- counts[, 3:ncol(counts)] # Remove gene ID columns
rownames(cleaned_data_matrix) <- counts$Geneid_clean # Add gene IDs as row names instead

# Make each treatment a factor (level)
samples$treatment <- factor(samples$treatment)

# Create a DGEList object
d <- DGEList(counts = as.matrix(cleaned_data_matrix),
                    group = samples$treatment)

# Filter out lowly expressed genes
keep <- filterByExpr(d, group = samples$treatment)
```

Trimmed Mean of M-values (TMM) normalisation

```
# Calculate scaling factors for normalisation through TMM
d <- calcNormFactors(d, method = "TMM")
```

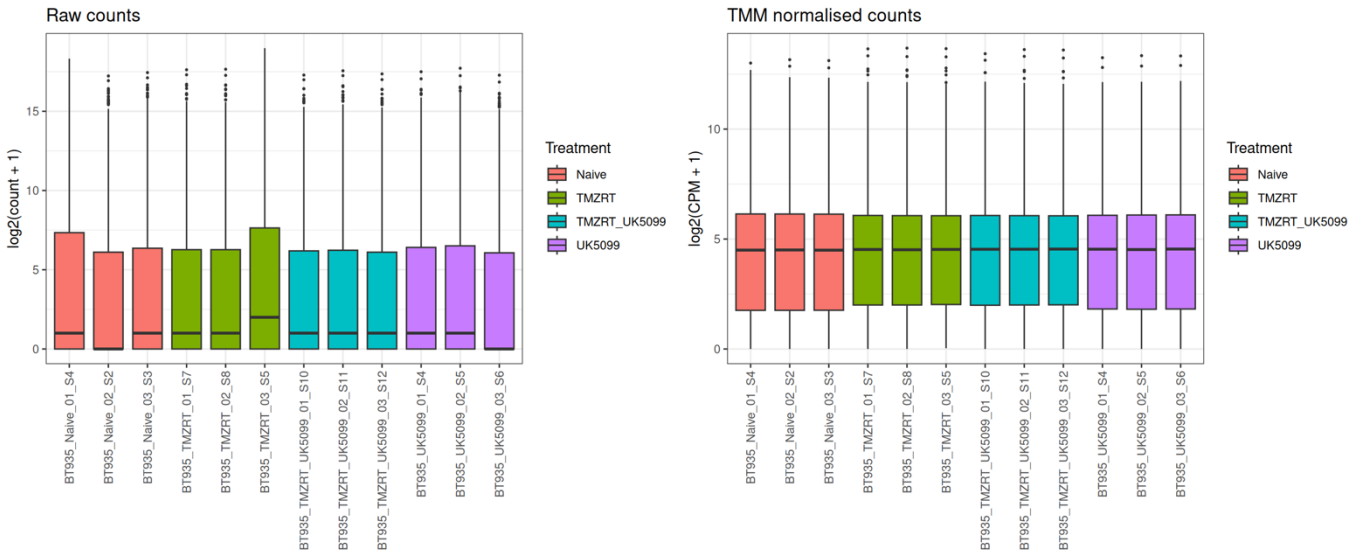


Figure 1: Boxplots of raw & TMM-normalised expression values across samples

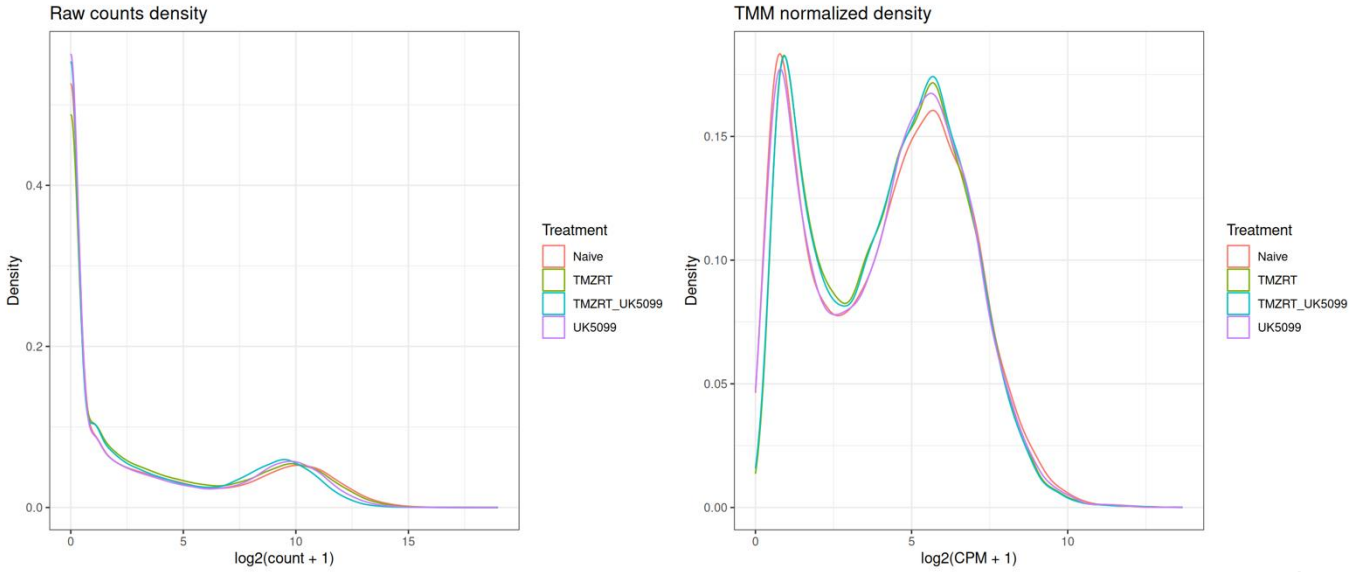


Figure 2: Density plots of raw & TMM-normalised expression distributions across genes

Data normalisation

edgeR: filterbyExpr(), calcNormFactors()

Differential expression analysis

```
qlf_combo_vs_tmzrt <- glmQLFTest(fit,
                                 contrast = contrasts[, "TMZRT_UK5099_vs_TMZRT"])
```

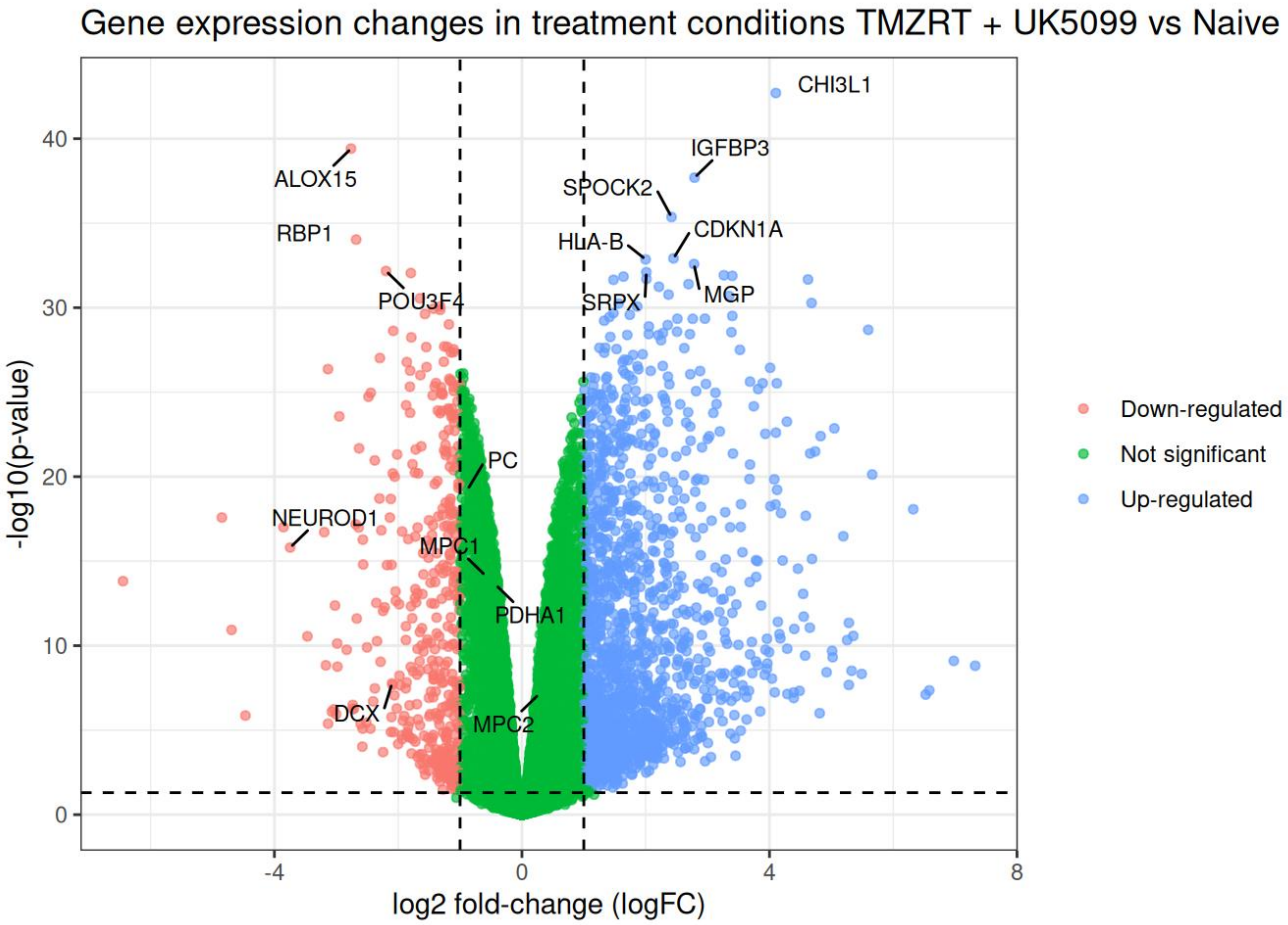


Figure 3: Volcano plot of differential expression results for combination treatment (TMZ-RT + UK5099) vs. Naive

Differential expression analysis

edgeR: glmQLFTest()

Thresholded over-representation analysis

Enriched in treated samples (up-regulated)

immune response, structural remodelling

p_value	term_size	term_name	source
0.0e+00	76	Interferon alpha/beta signaling	REAC
2.0e-07	74	cell adhesion mediated by integrin	GO:BP
7.0e-07	80	negative regulation of viral process	GO:BP
1.2e-06	55	negative regulation of viral genome replication	GO:BP
4.1e-06	89	Collagen formation	REAC

Enriched in naïve samples (down-regulated)

cell differentiation, communication/signalling

p_value	term_size	term_name	source
0.0108067	39	dorsal/ventral pattern formation	GO:BP
0.0108067	24	cell communication by electrical coupling involved in cardiac conduction	GO:BP
0.0160701	159	osteoblast differentiation	GO:BP
0.0177653	29	cell communication by electrical coupling	GO:BP
0.0180662	66	export across plasma membrane	GO:BP

All genes

protein synthesis, RNA processing, metabolic activity

p_value	term_size	term_name	source
0	163	cytoplasmic translation	GO:BP
0	161	mitochondrial gene expression	GO:BP
0	120	Eukaryotic Translation Initiation	REAC
0	120	Cap-dependent Translation Initiation	REAC
0	192	rRNA processing in the nucleus and cytosol	REAC

Thresholded over-representation analysis

g:Profiler (gprofiler2 R)

5 <= gene set size <= 200

GO:BP – version 2025-03-16
Reactome – version 2025-05-23
WikiPathways – version 2025-05-10

Non-thresholded gene set enrichment analysis

Enriched in treated samples

immune response

NAME	SIZE	NES	NOM p-val	FDR q-val
HALLMARK_INTERFERON_ALPHA_RESPONSE	78	2.650112	0	0.0000000
HALLMARK_INTERFERON_GAMMA_RESPONSE	128	2.495393	0	0.0000000
HALLMARK_TNFA_SIGNALING_VIA_NFKB	144	2.451210	0	0.0000000
INTERFERON ALPHA BETA SIGNALING	54	2.274128	0	0.0000000
CELLULAR RESPONSE TO TUMOR NECROSIS FACTOR	70	2.255014	0	0.0000000
HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	141	2.233496	0	0.0000000
INTERFERON GAMMA SIGNALING	63	2.208246	0	0.0000000
DEFENSE RESPONSE TO BACTERIUM	87	2.198123	0	0.0000000
HALLMARK_INFLAMMATORY_RESPONSE	118	2.168079	0	0.0002207
CYTOKINE CYTOKINE RECEPTOR INTERACTION	92	2.167530	0	0.0001986

Enriched in naïve samples

translation, protein synthesis

NAME	SIZE	NES	NOM p-val	FDR q-val
HALLMARK_MYC_TARGETS_V1	142	-2.933135	0	0
FORMATION OF A POOL OF FREE 40S SUBUNITS	89	-2.878914	0	0
CYTOPLASMIC RIBOSOMAL PROTEINS	77	-2.877303	0	0
REGULATION OF EXPRESSION OF SLITS AND ROBOS	138	-2.873212	0	0
EUKARYOTIC TRANSLATION INITIATION	106	-2.862629	0	0
CAP-DEPENDENT TRANSLATION INITIATION	106	-2.856724	0	0
GTP HYDROLYSIS AND JOINING OF THE 60S RIBOSOMAL SUBUNIT	99	-2.851567	0	0
NONSENSE-MEDIATED DECAY (NMD)	103	-2.847026	0	0
RRNA PROCESSING IN THE NUCLEUS AND CYTOSOL	174	-2.841860	0	0
MAJOR PATHWAY OF RRNA PROCESSING IN THE NUCLEOLUS AND CYTOSOL	165	-2.841448	0	0

Non-thresholded gene set enrichment analysis

GSEA (command-line)

5 <= gene set size <= 200

Bader lab – GO:BP pathways, version 2025-09-01

Enrichment map

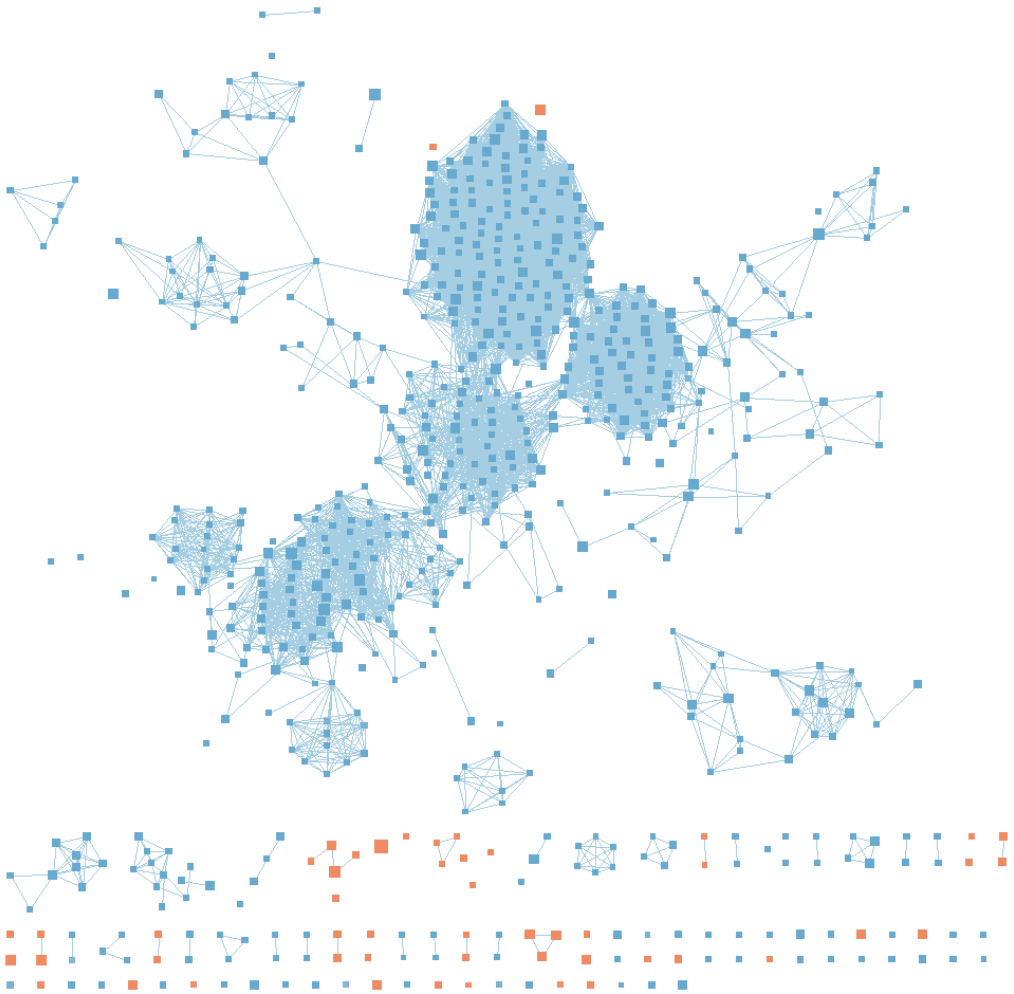


Figure 4. Initial enrichment map of GSEA results for the TMZRT + UK5099 vs Naive comparison

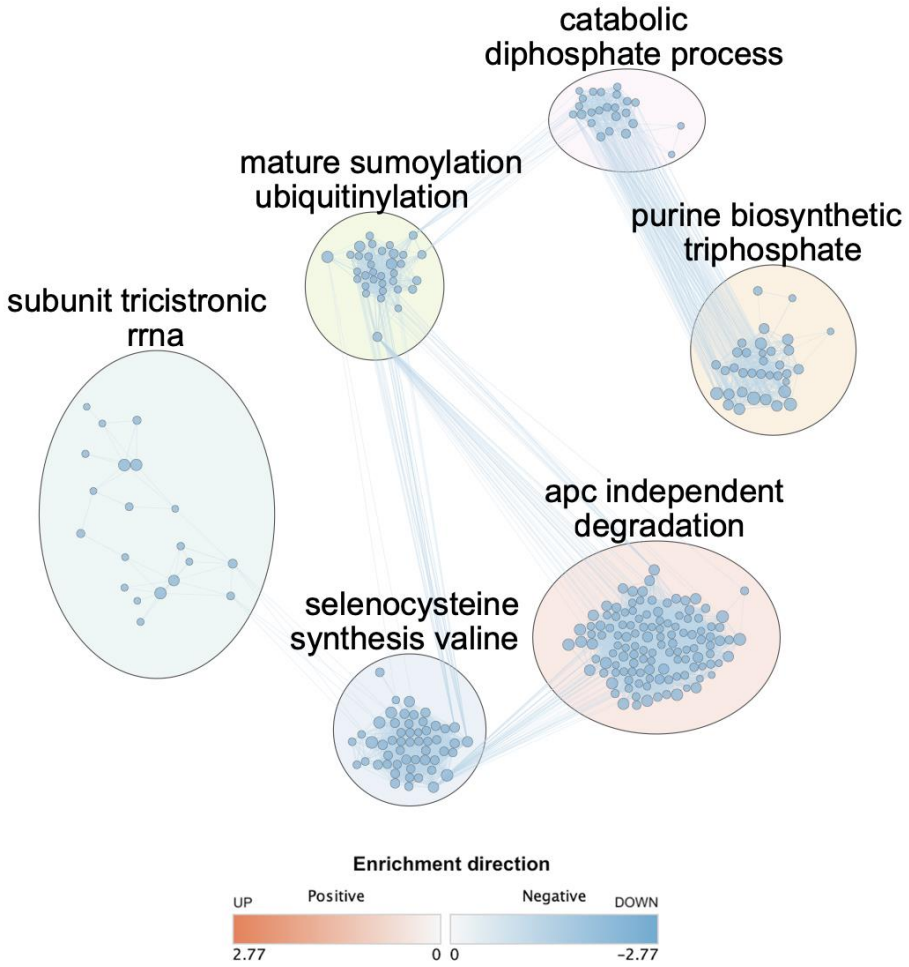


Figure 5. Annotated enrichment map summarizing major pathway themes identified from the GSEA analysis of the TMZRT + UK5099 vs Naive comparison

Enrichment map
Cytoscape

Key insights

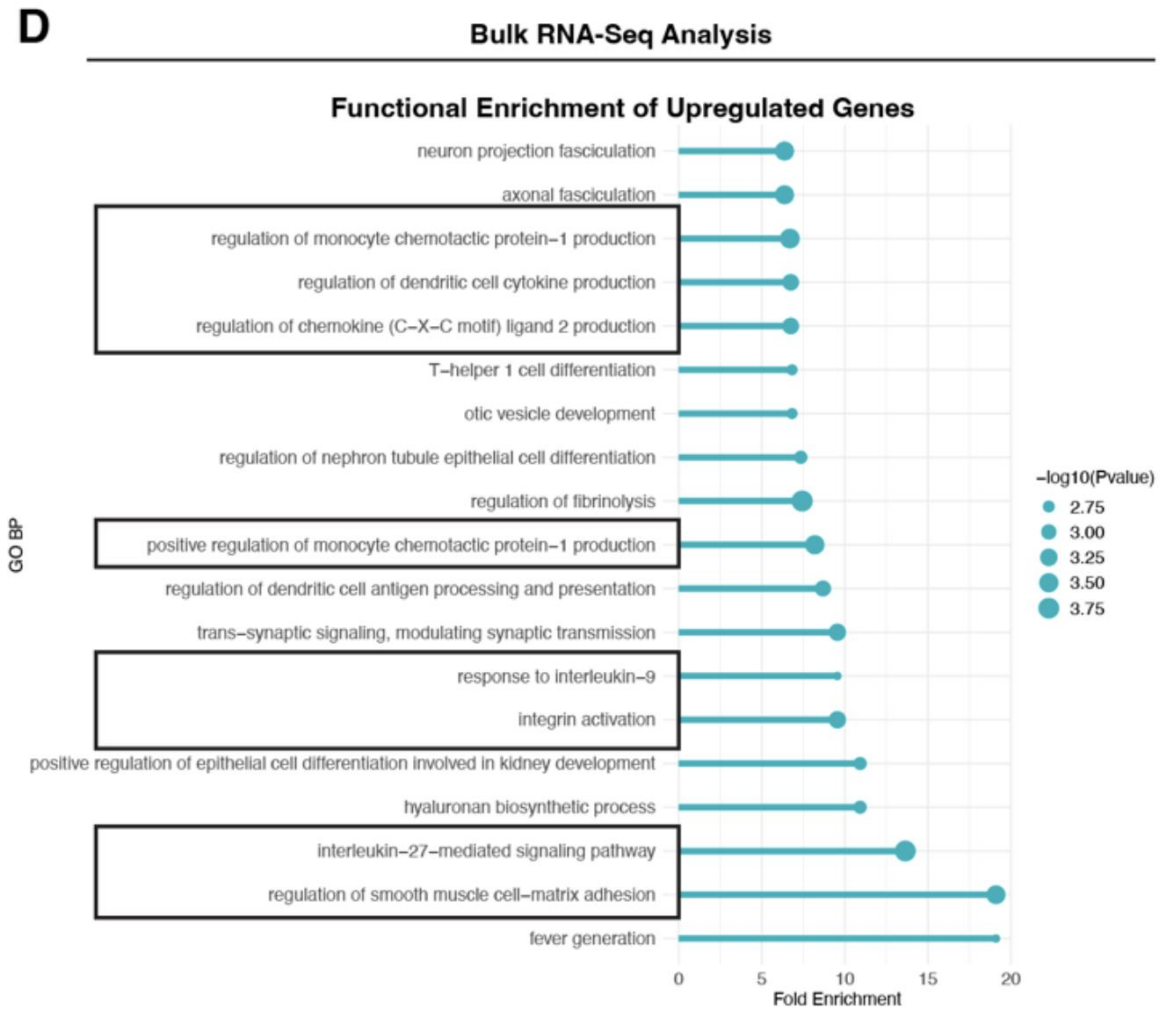


Figure 4. Supplementary Figure S4D from the paper "Disease stage-specific role of the mitochondrial pyruvate carrier suppresses differentiation in temozolomide and radiation-treated glioblastoma" (Martell et al., 2025): Functional enrichment of upregulated genes from bulk RNA-seq analysis

Up-regulated genes

- Groups: immune response, structural remodelling
- Matches the original paper's enrichment results
  - Interleukin response + fever generation → immune response
- Dying cancer cells → trigger DAMPs → release cytokines → immune response (Hänggi & Ruffell, 2023)
- Therapy → ECM remodelling → protect tumour from anti-cancer agents (Prakash & Shaked, 2024)

Down-regulated genes

- Groups: cell differentiation, protein synthesis
- GSCs → osteoblasts (bone) → increases tumour growth and invasion (Ricci-Vitiani et al., 2008)
- Overexpression of certain proteins → increases tumour growth and invasion (Cornelissen et al., 2024)

Thank you!